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Correlating retinal and choroidal vascular parameters with volumetric MRI in Alzheimer's disease and amnestic mild cognitive impairment

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Abstract

Background We assess the relationships between retinal and choroidal structural and microvascular parameters and brain volumetric magnetic resonance imaging (MRI) parameters in individuals with amnestic mild cognitive impairment (aMCI) and Alzheimer's disease (AD).

Methods Participants underwent optical coherence tomography (OCT) and OCT angiography (OCTA) imaging using the Zeiss Cirrus HD-5000 AngioPlex (Carl Zeiss Meditec, Dublin, CA) and volumetric brain MRI imaging with NeuroQuant (CorTechs Labs, San Diego, CA) analysis. Retinal and choroidal structural and microvascular parameters were extracted from OCT and OCTA scans. Superficial capillary plexus perfusion density and vessel density in the 3 × 3 mm and 6 × 6 mm circles and rings centered on the fovea were determined. Hippocampal, superior lateral ventricle (SLV), and inferior lateral ventricle (ILV) volumes were determined. Correlations between OCT, OCTA, and volumetric MRI parameters in patients with aMCI and AD were explored using Spearman partial correlation analysis.

Results 68 eyes of 37 aMCI participants and 64 eyes of 33 AD participants were analyzed. In the eyes with aMCI, hippocampal volume negatively correlated with FAZ area and positively correlated with perfusion density and vessel density in the 3 mm OCTA scan. In the aMCI cohort, SLV and ILV volume inversely correlated with perfusion density in the 3 mm and 6 mm OCTA scans, respectively. In the eyes with AD, SLV and ILV volume inversely correlated with perfusion density and vessel density in the 3 mm and 6 mm OCTA scans. Central subfield thickness, ganglion cell-inner plexiform layer thickness, RNFL thickness, and choroidal vascularity index did not significantly correlate with SLV, ILV, or hippocampal volume in any cohorts ($p > 0.05$).

Conclusions Decline in retinal microvascular parameters significantly correlated with hippocampal volume loss and ventricular expansion, suggesting that these parameters may mirror cerebral neurodegeneration in individuals with aMCI and AD.

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Keywords OCT, OCT angiography, Volumetric MRI, Alzheimer's disease, Mild cognitive impairment

Background

Alzheimer's disease (AD) is a leading cause of disability worldwide, with the number of individuals developing dementia increasing in the setting of population growth and aging [1]. While FDA approved medications including lecanemab and donanemab may decrease amyloid-beta burden and slow the rate of cognitive loss, there is currently no cure for AD [2, 3]. Additionally, early diagnosis and lifestyle modifications may slow the progression of AD symptoms in susceptible individuals [4, 5, 6]. As such, an increasing focus has been placed on identifying novel biomarkers for early identification of AD [7]. Plasma biomarkers have become increasingly promising, as they may correlate with the presence of amyloid on positron emission tomography (PET) imaging and have reasonable discriminative capacity between AD and other forms of dementia [8, 9, 10]. As we now define AD biologically rather than syndromically, biomarkers will play an ever increasing role and become more prevalent [11]. It will be essential to develop non-invasive, less expensive biomarkers for use in screening as well as in targeted assessment of at-risk and symptomatic individuals.

Retinal imaging has been explored as a source of novel biomarkers for neurodegenerative disease. We previously demonstrated that individuals with AD have decreased retinal vessel density (VD) and perfusion density (PD) compared to cognitively normal individuals [12]. Promising data also suggests that retinal changes may precede symptom onset in AD-susceptible individuals [13, 14]. Hippocampal atrophy on magnetic resonance imaging (MRI) has been associated with the pathophysiology of AD and has been shown to precede symptom onset in AD [15]. The use of volumetric MRI to quantify three-dimensional volumes of brain structures in the medial temporal lobe, including the amygdala, hippocampus, and parahippocampal gyrus, has also been shown to discriminate among individuals with normal cognition, mild cognitive impairment (MCI), and AD [16]. Technological advances in MRI analysis have allowed for a fully-automated software package to quantify brain volumes called NeuroQuant (CorTechs Labs, San Diego, CA) [17]. Research using NeuroQuant has shown that hippocampal volume may be a strong predictor of conversion from MCI to AD [18].

Our group previously published a pilot study correlating optical coherence tomography angiography (OCTA) parameters and NeuroQuant volumetric MRI parameters in individuals with MCI and AD [19]. This study showed that while OCTA parameters did not significantly correlate with hippocampal or superior lateral ventricle (SLV) volumes, VD in the 6 mm circle and 3 mm ring and PD in

the 3 mm ring had a significant negative correlation with inferior lateral ventricle (ILV) volume. To learn more, we have incorporated a larger patient cohort which includes the individuals from this previously published cohort. In addition, we have added choroidal vascularity index (CVI), a parameter recently shown to differ between AD and controls [20]. In this paper, we aim to more definitively characterize how retinal and choroidal microvascular and structural parameters may correlate with volumetric MRI in a population of individuals with aMCI and AD.

Methods

This cross-sectional study (NCT03233646) was reviewed and approved by the Duke University Health System Institutional Review Board (Pro00111831). Written informed consent was obtained from all participants and/or their designated legally authorized representative prior to study enrollment. This study adhered to the tenets of the Declaration of Helsinki with regard to enrollment of human subjects.

Individuals 50 years or older with a diagnosis of aMCI or AD were identified and recruited from the Duke Neurology Memory Disorders Clinic in Durham, North Carolina. Standardized criteria established by the National Institute on Aging and the Alzheimer's Association were used to define MCI, which included: "concern regarding a change in cognition, objective evidence of impairment in one or more cognitive domain, preservation of independence in functional abilities, and not demented." [21]. Clinical assessment included neurocognitive testing (e.g., Montreal Cognitive Assessment [MoCA]) and brain MRI. A diagnosis of aMCI was assigned if memory impairment was determined to be the initial and most prominent cognitive deficit on MoCA. All diagnoses were reviewed and confirmed by expert clinicians experienced in memory disorders (AJL, KGJ) in accordance with published guidelines from the National Institute on Aging/Alzheimer's Association [7, 21, 22]. Individuals with subjective cognitive impairment only or a diagnosis of dementia unrelated to AD were excluded from enrollment. Positron emission tomography imaging and cerebrospinal fluid (CSF) sampling were not performed as part of standard of care at our institution. Exclusion criteria for this study also included diabetes mellitus, uncontrolled hypertension, demyelinating disorders, glaucoma or glaucoma suspect, vitreoretinal pathology, refractive error $\pm$ 6 diopters or more, and/or corrected visual acuity worse than 20/40 at the time of ophthalmic imaging. Ocular history was taken from study participants and their caregiver(s) and using electronic health record

data from available ophthalmic care. Ultra-widefield fundus images (California; Optos, Marlborough, MA) were acquired on all study participants and reviewed by study staff for ophthalmic pathology, and any such individuals were excluded from statistical analysis. The Mini-Mental State Examination (MMSE) was performed on all study participants at the time of image acquisition, and years of education were reported from the first grade onwards.

Retinal image acquisition

All study participants were imaged using the Zeiss Cirrus HD-5000 Spectral-Domain OCT with AngioPlex (Carl Zeiss Meditec, Dublin, CA, Software Version 11.0.0.29946). 3×3 mm and 6×6 mm OCTA images centered on the fovea were acquired. For the 3×3 mm OCTA scans, 245 A-lines formed one B-scan, while B-scans were acquired at 245 different spatial locations and repeated four times at each location, resulting in a spacing of $12.2\text{ }\mu\text{m}$ between pixels in the en face OCTA images. For the 6×6 mm OCTA scans, 350 A-lines formed one B-scan while B-scans were acquired at 350 different spatial locations and repeated twice at each location resulting in a spacing of $17.1\text{ }\mu\text{m}$ between pixels in the en face OCTA images. The scan speed for OCTA was 68,000 A-scans/second with A-scan depth of 2 mm. OCT and OCTA images of poor quality (identified on manual review), such as less than 7/10 signal strength or with low spatial resolution, poor centration around the fovea, segmentation errors, projection artifact, and/or motion artifact, were excluded prior to statistical analysis. We excluded 8 eyes total due to these factors. Embedded AngioPlex software was used to perform automated segmentation and analysis of the superficial capillary plexus (SCP). The software quantified average SCP VD and PD in the 3 mm circle and ring and in the 6 mm circle, outer ring, and inner ring using the Early Treatment Diabetic Retinopathy Study (ETDRS) grid overlay. Vessel density was defined as the total length of perfused vasculature in a given unit area (mm^{-1}), whereas PD was defined as the total area of perfused vasculature in a given unit area (unitless ratio). The foveal avascular zone (FAZ) was automatically segmented and quantified from 3×3 mm OCTA scans using the AngioPlex software.

OCT image acquisition included a $200\times 200\text{ }\mu\text{m}$ optic disc cube, a 512×128 macular cube, and a 21-line enhanced depth imaging raster scan. The optic disc cube scan contained 200 horizontal B-scans comprising 200 A-scan per B-scan within a cube measuring $6\times 6\times 2$ mm centered at optic disc center. The macular cube scan consisted of a 6 mm square grid with 128 horizontal scan lines each composed of 512 A-scans using a 47 mm spacing between lines and 1024 data points per A-scan. In the macular cube scan, a 14.13 mm^2 elliptical annulus area that was centered on the fovea was used to

quantify the average GC-IPL thickness. In the macular cube scan, CST was calculated as the distance between the retinal pigment epithelium and the inner limiting membrane at the fovea on the macular cube scan. In the optic disc cube scan, average thickness of the RNFL was measured in a 3.46 mm diameter circle centering on the optic disc. Enhanced depth imaging (EDI) foveal OCT scans were acquired using the HD21 line scan protocol with 21 horizontal scan lines 9 mm in length, at a depth of 2 mm with 8 acquired B-scans for each line and each B-scan composed of 1024 A-scans. Images were captured with the EDI mode so that the best signal-to-noise ratio was obtained at the bottom of the B-scan. Semi-automated segmentation and binarization of the choroid was performed using Comprehensive Ocular Imaging Network (COIN) (www.ocularimaging.net) software [23]. From these segmented images, total choroidal area (TCA) and luminal area (LA) were calculated, and the ratio of these variables (LA/TCA) was defined as CVI.

Volumetric MRI acquisition and analysis

All subjects underwent non-contrast brain MRI using Siemens 3 Tesla Skyra with sagittal 3D T1-weighted scans and a 20-channel head coil. MRI acquisition parameters include a field of view (FOV) of 256 mm and slice thickness of 1.2 mm. Images were processed using NeuroQuant version 2.3.0 (CorTechs Labs, San Diego, CA), and volumes of regional brain structures were included in the Age Related Atrophy Report [17]. Volume morphometry was available for hippocampal volume, inferior lateral ventricle volume, and superior lateral ventricle volume. Total volume in cubic centimeters (cm^3) was determined for each brain region.

Statistical analysis

In cases where both eyes of the same patient were imaged, the two values for each retinal imaging parameter were averaged. Wilcoxon rank sum tests were performed for the distributions of each OCT, OCTA, and MRI parameter between patients with AD and aMCI. Spearman partial correlation analysis was performed to explore the relationships between each OCT or OCTA parameter and each MRI parameter while adjusting for age, sex, race, and treated hypertension. The level of significance was set at $p=0.05$. Due to the exploratory nature of this pilot study, no adjustment was made for multiple comparisons.

Results

132 eyes (68 aMCI and 64 AD) from 70 patients (37 aMCI and 33 AD) were analyzed. The mean age was 71.9 ± 7.0 years for the aMCI group and 75.8 ± 7.5 years for the AD group ($p=0.027$). The aMCI group had 49% females ($n=18/37$), and the AD group had 64% females

($n = 21/33$) (Table 1). No significant difference in years of education was observed between the groups (15.7 ± 2.4 for aMCI vs. 15.8 ± 2.8 for AD). The mean MMSE score was 25.1 ± 4.9 in the aMCI group and 22.4 ± 5.7 in the AD group ($p = 0.033$). Mean OCT, OCTA, and MRI volumetric parameters as well as p values for Wilcoxon rank sum tests are provided in Table 2. Spearman partial correlation coefficients and p values for all correlations are provided in Tables 3 and 4. Figure 1 illustrates two representative examples of NeuroQuant brain MRI reports with corresponding brain volumes.

The mean 3 mm circle VD was 18.29 ± 2.31 for the AD group and 19.62 ± 1.72 for the aMCI group ($p = 0.018$) (Table 2). The mean 3 mm ring VD was 19.29 ± 2.22 for the AD group and 20.66 ± 1.72 for the aMCI group ($p = 0.013$). The mean GC-IPL thickness was 72.14 ± 7.18 for the AD group and 75.57 ± 7.25 for the aMCI group ($p = 0.045$). The mean ILV volume was 3.72 ± 1.77 for the AD group and 2.95 ± 1.35 for the aMCI group ($p = 0.042$). There were no significant differences between cohorts in other OCT, OCTA, or volumetric MRI parameters.

Correlation analysis

Amnesic MCI cohort

In patients with aMCI ($n = 37$), hippocampal volume positively correlated with PD in the 3 mm circle ($\rho = 0.4837$, $p = 0.0058$) (Table 3). Hippocampal volume also positively correlated with VD in the 3 mm circle ($\rho = 0.4449$, $p = 0.0122$) and ring ($\rho = 0.3842$, $p = 0.0329$). Hippocampal volume negatively correlated with FAZ area ($\rho = -0.3958$, $p = 0.0275$). SLV volume negatively correlated with PD in

the 3 mm circle ($\rho = -0.4232$, $p = 0.0177$). ILV negatively correlated with PD in the 6 mm inner ring ($\rho = -0.3585$, $p = 0.0476$).

AD cohort

In AD patients ($n = 33$), ILV volume negatively correlated with PD in the 3 mm circle ($\rho = -0.4515$, $p = 0.0140$) and ring ($\rho = -0.4824$, $p = 0.0081$), and the 6 mm circle ($\rho = -0.3749$, $p = 0.0451$) and inner ring ($\rho = -0.4246$, $p = 0.0217$) (Table 4). ILV volume also negatively correlated with VD in the 3 mm circle ($\rho = -0.4123$, $p = 0.0262$) and ring ($\rho = -0.4812$, $p = 0.0082$), and 6 mm inner ring ($\rho = -0.4161$, $p = 0.0248$). SLV volume negatively correlated with PD in the 6 mm circle ($\rho = -0.3792$, $p = 0.0425$). SLV volume also negatively correlated with VD in the 3 mm ring ($\rho = -0.3786$, $p = 0.0429$), and in the 6 mm circle ($\rho = -0.4052$, $p = 0.0292$) and outer ring ($\rho = -0.4297$, $p = 0.0200$). There were no significant correlations between ILV, SLV, or hippocampal volume and CST, CVI, RNFL thickness, and GC-IPL thickness ($p > 0.05$).

Discussion

We investigated the relationships between retinal imaging parameters and brain volumetric MRI measurements in individuals with aMCI and AD. In our group's previous work, Yoon et al. correlated OCT and OCTA parameters and volumetric MRI measurements in a smaller combined cohort comprising 9 AD and 7 MCI patients [19]. We found significant correlations between ILV volume and VD and PD in the 6 mm inner ring and VD in the 6 mm circle. With the increased sample size herein and no Bonferroni correction due to this being a hypothesis-generating exploratory study, we performed correlation analyses in aMCI and AD patients as separate cohorts and identified a broader range of significant correlations. In this study, we found that reductions in PD in the 3 mm and 6 mm OCTA images significantly correlated with SLV and ILV volume expansion in aMCI patients. Additionally, reductions in FAZ area and VD and PD in the 3 mm OCTA images significantly correlated with decreased hippocampal volume in aMCI patients. Meanwhile, reductions in VD and PD in the 3 mm and 6 mm OCTA images significantly correlated with ILV and SLV volume expansion in AD patients.

In the aMCI cohort, SLV volume expansion significantly correlated with decreasing PD in the 3 mm circle, and ILV volume expansion significantly correlated with decreasing PD in the 6 mm inner ring. Previous studies have reported brain volume changes in patients with MCI. Evans et al. reported that MCI patients who progressed to AD exhibited significantly higher rates of whole brain volume atrophy and ventricular expansion compared to clinically stable MCI patients [24]. Clifford et al. reported that whole brain atrophy and ventricular

Table 1 Demographic information and clinical characteristics of patients with Alzheimer's disease and amnesic MCI

Variable	Statistic	AD	Amnesic MCI	P-value*
Age	N	33	37	0.027
	Mean (SD)	75.78 (7.53)	71.86 (6.97)	
	Min, Median, Max	58.1, 77.3, 88.8	52.1, 73.6, 82.7	
Sex	N (%)			0.208
Male		12 (36.4)	19 (51.4)	
Female		21 (63.6)	18 (48.6)	
MMSE Score	N	32	33	0.033
	Mean (SD)	22.38 (5.72)	25.12 (4.91)	
	Min, Median, Max	7.0, 23.5, 30.0	8.0, 27.0, 30.0	
Years of Education	N	32	36	0.894
	Mean (SD)	15.75 (2.76)	15.67 (2.38)	
	Min, Median, Max	12.0, 16.0, 22.0	12.0, 16.0, 20.0	

*P-values based on t-test for continuous variables and Chi-square for sex

AD = Alzheimer's disease; mci = mild cognitive impairment; mmse = mini mental state examination

Table 2 OCT and OCTA parameters for subjects with Alzheimer's disease and amnestic MCI

Variable	Statistic	AD	Amnestic MCI	P Value
FAZ Area	N	33	35	0.659
	Mean (SD)	0.22 (0.13)	0.21 (0.07)	
	Min, Median, Max	0.1, 0.2, 0.7	0.1, 0.2, 0.3	
3 mm Circle Perfusion Density	N	33	37	0.066
	Mean (SD)	0.34 (0.04)	0.35 (0.03)	
	Min, Median, Max	0.3, 0.4, 0.4	0.3, 0.4, 0.4	
3 mm Ring Perfusion Density	N	33	37	0.782
	Mean (SD)	0.35 (0.04)	0.34 (0.07)	
	Min, Median, Max	0.3, 0.4, 0.4	0.2, 0.4, 0.4	
3 mm Circle Vessel Density	N	33	37	0.018*
	Mean (SD)	18.29 (2.31)	19.62 (1.72)	
	Min, Median, Max	13.2, 18.9, 22.1	16.1, 19.9, 22.7	
3 mm Ring Vessel Density	N	33	37	0.013*
	Mean (SD)	19.29 (2.22)	20.66 (1.72)	
	Min, Median, Max	14.4, 20.0, 22.5	16.6, 20.9, 23.6	
6 mm Circle Perfusion Density	N	33	37	0.246
	Mean (SD)	0.41 (0.05)	0.42 (0.04)	
	Min, Median, Max	0.2, 0.4, 0.5	0.3, 0.4, 0.5	
6 mm Outer Ring Perfusion Density	N	33	37	0.174
	Mean (SD)	0.42 (0.05)	0.43 (0.04)	
	Min, Median, Max	0.2, 0.4, 0.5	0.3, 0.4, 0.5	
6 mm Inner Ring Perfusion Density	N	33	37	0.444
	Mean (SD)	0.40 (0.06)	0.42 (0.04)	
	Min, Median, Max	0.2, 0.4, 0.5	0.2, 0.4, 0.5	
6 mm Circle Vessel Density	N	33	37	0.295
	Mean (SD)	16.79 (1.94)	17.27 (1.56)	
	Min, Median, Max	9.7, 17.4, 19.3	11.0, 17.6, 19.2	
6 mm Outer Ring Vessel Density	N	33	37	0.158
	Mean (SD)	17.11 (1.86)	17.61 (1.52)	
	Min, Median, Max	9.9, 17.7, 19.5	11.4, 17.9, 19.4	
6 mm Inner Ring Vessel Density	N	33	37	0.365
	Mean (SD)	16.71 (2.41)	17.44 (1.60)	
	Min, Median, Max	10.0, 17.5, 20.0	10.7, 17.9, 19.6	
Central Subfield Thickness	N	33	37	0.588
	Mean (SD)	263.06 (24.41)	266.55 (25.42)	
	Min, Median, Max	201.5, 259.0, 316.0	215.0, 266.5, 333.5	
Choroidal Vascularity Index	N	33	37	0.495
	Mean (SD)	0.64 (0.02)	0.64 (0.01)	
	Min, Median, Max	0.6, 0.6, 0.7	0.6, 0.6, 0.7	
Ganglion Cell-Inner Plexiform Layer	N	33	37	0.045*
	Mean (SD)	72.14 (7.18)	75.57 (7.25)	
	Min, Median, Max	57.5, 72.0, 86.5	56.0, 75.0, 92.5	
Retinal Nerve Fiber Layer	N	33	37	0.400
	Mean (SD)	86.53 (9.29)	88.23 (9.94)	
	Min, Median, Max	72.0, 84.0, 107.5	69.0, 91.0, 112.0	
Hippocampal Volume	N	33	37	0.715
	Mean (SD)	5.93 (1.28)	6.03 (1.28)	
	Min, Median, Max	3.4, 5.8, 8.8	3.1, 6.1, 9.2	
Superior Lateral Ventricle Volume	N	33	37	0.518
	Mean (SD)	49.06 (23.77)	45.96 (21.74)	
	Min, Median, Max	12.1, 44.5, 122.6	14.0, 40.3, 115.4	
Inferior Lateral Ventricle Volume	N	33	37	0.042*

Table 2 (continued)

Variable	Statistic	AD	Amnesic MCI	P Value
	Mean (SD)	3.72 (1.77)	2.95 (1.35)	
	Min, Median, Max	0.9, 3.7, 9.1	1.3, 2.6, 6.4	

* P value significant as < 0.05

OCTA = optical coherence tomography angiography; MRI = magnetic resonance imaging; AD = Alzheimer's disease; MCI = Mild cognitive impairment; CST = central subfield thickness; CVI = choroidal vascularity index; GC-IPL = ganglion cell–inner plexiform layer; RNFL = retinal nerve fiber layer; FAZ = foveal avascular zone

expansion rates were accelerated in aMCI patients who were APOE $\epsilon 4$ carriers compared to non-carriers [25]. Patients with aMCI have demonstrated shared amyloid-beta pathology with AD on PET imaging as well as higher rates of conversion to AD compared to non-amnesic MCI [26, 27]. The observed correlation between reduced retinal PD and ventricular expansion may reflect an underlying neurodegenerative process affecting both the cerebral and retinal microvasculature and suggests potential of retinal measurements to be used as a surrogate.

In the aMCI cohort, decreasing hippocampal volume significantly correlated with decreasing PD in the 3 mm circle, as well as with decreasing VD in the 3 mm circle and 3 mm ring. Multiple studies have reported hippocampal atrophy in aMCI. Mueller et al. found that patients with MCI had smaller CA1 and CA2 hippocampal subfield volumes compared to controls [28]. Emmert et al. and Csukly et al. observed significantly lower hippocampal volumes in aMCI patients compared to non-amnesic MCI patients [29, 30]. Several studies have also compared OCTA parameters in MCI patients and controls. Criscuolo et al. reported that VD was significantly reduced in multiple-domain aMCI patients compared to controls [31]. Likewise, Zhang et al. found that parafoveal superficial capillary plexus VD and adjusted flow index (AFI) were significantly lower in aMCI and early AD patients than in controls [32]. These findings suggest that in aMCI patients who are at risk of progressing to AD, microvasculature damage contributes to hippocampal atrophy and reduced retinal vessel and perfusion density.

In this study, hippocampal volume negatively correlated with FAZ area in patients with aMCI. The FAZ is bordered by parafoveal capillaries, and its enlargement may be attributed to capillary dropout [33]. This may reflect underlying neurodegenerative changes secondary to amyloid- β deposition within the retinal microvasculature. In patients where decreased hippocampal volume is significantly correlated with reduced parafoveal vessel density, it would follow that FAZ may also be enlarged. Several studies have reported increased FAZ area in patients with MCI and AD. Jiang et al. found significantly larger FAZ areas and decreased superficial capillary plexus vessel density in both MCI and AD patients compared to controls [34]. Similarly, Bulut et al. reported increased FAZ and decreased vessel density in

AD patients compared to controls [35]. In a case-control study by O'Bryhim et al., individuals with preclinical AD confirmed by the presence of amyloid- β biomarker also had significantly larger FAZ areas than individuals without amyloid- β biomarker [13]. Criscuolo et al. reported that FAZ area was increased in aMCI patients compared to controls, and was one of the most sensitive vascular parameters that distinguished both single-domain and multiple-domain MCI patients from controls [31]. In contrast, other studies such as those by Yoon et al. and Zhang et al. found no significant difference in FAZ area among MCI, AD, and control groups [32, 36]. These conflicting findings suggest ongoing variability in the literature and the need for further investigation into the role of FAZ as a biomarker in neurocognitive disease.

In the AD cohort, ILV and SLV volume expansion significantly correlated with decreasing VD and PD in the 3 mm and 6 mm OCTA images. In previous studies, ventricular volume expansion in AD has been correlated with cognitive decline [37, 38]. Ventricular expansion is well-characterized in AD and is thought to be driven by medial temporal lobe and whole brain atrophy. One proposed mechanism is the accumulation of pathologic amyloid-beta plaques in brain tissue as well as cerebral small vessels, as seen in cerebral amyloid angiopathy. Among all OCT and OCTA parameters, only VD and PD significantly correlated with volumetric MRI parameters, suggesting that OCTA may be more sensitive to AD-related brain atrophy than OCT structural measures.

In the AD cohort, there were no significant correlations between hippocampal volume and OCT or OCTA parameters. This absence of significant correlations may reflect the well-documented trajectory of hippocampal atrophy in AD wherein hippocampal volume loss follows a nonlinear, sigmoidal trajectory characterized by an initial acceleration of atrophy followed by a deceleration phase [39]. AD-related cortical thinning is most rapid when MMSE scores are in the range of 20–23 corresponding to mild dementia and slows thereafter. Our AD cohort (average MMSE of 22.4) falls within this deceleration phase, though slight deviations may arise due to population differences. Therefore, it is possible that in our AD cohort, hippocampal atrophy reached a plateau, and further changes to brain volume were driven by atrophy in other brain regions.

Table 3 Spearman correlation coefficients of OCT, OCTA, and volumetric brain MRI parameters for amnesic MCI patients

Parameter	Spearman ρ	P Value
Hippocampal Volume	0.2000	0.2808
CST	0.0414	0.8252
CVI	-0.0221	0.9063
RNFL Thickness	0.1467	0.4309
GC-IPL Thickness	-0.3958	0.0275*
FAZ Area		
6 mm Circle PD	0.2180	0.2388
6 mm Inner Ring PD	0.2014	0.2773
6 mm Outer Ring PD	0.2123	0.2516
3 mm Circle PD	0.4837	0.0058*
3 mm Ring PD	0.0461	0.8054
6 mm Circle VD	0.0899	0.6307
6 mm Inner Ring VD	0.1744	0.3481
6 mm Outer Ring VD	0.2252	0.2231
3 mm Circle VD	0.4449	0.0122*
3 mm Ring VD	0.3842	0.0329*
Superior Lateral Ventricle Volume	-0.0004	0.9981
CST	-0.0128	0.9456
CVI	0.1548	0.4057
RNFL Thickness	0.0801	0.6684
GC-IPL Thickness	0.0313	0.8672
FAZ Area		
6 mm Circle PD	-0.2709	0.1405
6 mm Inner Ring PD	-0.3458	0.0567
6 mm Outer Ring PD	-0.2737	0.1362
3 mm Circle PD	-0.4232	0.0177*
3 mm Ring PD	-0.1629	0.3814
6 mm Circle VD	-0.1810	0.3299
6 mm Inner Ring VD	-0.3160	0.0833
6 mm Outer Ring VD	-0.3035	0.0970
3 mm Circle VD	-0.3072	0.0928
3 mm Ring VD	-0.3462	0.0564
Inferior Lateral Ventricle Volume	-0.1386	0.4571
CST	0.0505	0.7874
CVI	0.2408	0.1919
RNFL Thickness	0.1010	0.5888
GC-IPL Thickness	0.0413	0.8253
FAZ Area		
6 mm Circle PD	-0.2700	0.1419
6 mm Inner Ring PD	-0.3585	0.0476*
6 mm Outer Ring PD	-0.2072	0.2634
3 mm Circle PD	-0.3255	0.0739
3 mm Ring PD	-0.0801	0.6686
6 mm Circle VD	-0.0827	0.6583
6 mm Inner Ring VD	-0.3407	0.0607
6 mm Outer Ring VD	-0.1860	0.3164
3 mm Circle VD	-0.2164	0.2423
3 mm Ring VD	-0.2299	0.2135

* P value significant as < 0.05

OCTA=optical coherence tomography; OCTA=optical coherence tomography angiography; MRI=magnetic resonance imaging; CST=central subfield thickness; CVI=choroidal vascularity index; GC-IPL=ganglion cell- inner plexiform layer; RNFL=retinal nerve fiber layer; FAZ=foveal avascular zone; PD=perfusion density; VD=vessel density

Table 4 Spearman correlation coefficients of OCT, OCTA, and volumetric brain MRI parameters for Alzheimer's disease patients

Parameter	Spearman ρ	P Value
Hippocampal Volume	-0.1210	0.5320
CST	0.2597	0.1737
CVI	-0.0749	0.6994
RNFL Thickness	0.1015	0.6003
GC-IPL Thickness	0.2258	0.2389
FAZ Area		
6 mm Circle PD	-0.0559	0.7733
6 mm Inner Ring PD	0.0299	0.8775
6 mm Outer Ring PD	-0.0153	0.9373
3 mm Circle PD	-0.2570	0.1783
3 mm Ring PD	-0.2330	0.2239
6 mm Circle VD	-0.0317	0.8703
6 mm Inner Ring VD	0.0259	0.8940
6 mm Outer Ring VD	-0.0581	0.7649
3 mm Circle VD	-0.2933	0.1226
3 mm Ring VD	-0.2872	0.1309
Superior Lateral Ventricle Volume	0.0894	0.6445
CST	-0.2133	0.2665
CVI	-0.0987	0.6104
RNFL Thickness	-0.0050	0.9794
GC-IPL Thickness	-0.0356	0.8546
FAZ Area		
6 mm Circle PD	-0.3792	0.0425*
6 mm Inner Ring PD	-0.2958	0.1193
6 mm Outer Ring PD	-0.3533	0.0601
3 mm Circle PD	-0.2961	0.1189
3 mm Ring PD	-0.3225	0.0879
6 mm Circle VD	-0.4052	0.0292*
6 mm Inner Ring VD	-0.3274	0.0830
6 mm Outer Ring VD	-0.4297	0.0200*
3 mm Circle VD	-0.2877	0.1301
3 mm Ring VD	-0.3786	0.0429*
Inferior Lateral Ventricle Volume	-0.0010	0.9958
CST	-0.3303	0.0801
CVI	0.1796	0.3511
RNFL Thickness	0.0442	0.8197
GC-IPL Thickness	0.1279	0.5084
FAZ Area		
6 mm Circle PD	-0.3749	0.0451*
6 mm Inner Ring PD	-0.4246	0.0217*
6 mm Outer Ring PD	-0.2769	0.1460
3 mm Circle PD	-0.4515	0.0140*
3 mm Ring PD	-0.4824	0.0081*
6 mm Circle VD	-0.3628	0.0531
6 mm Inner Ring VD	-0.4161	0.0248*
6 mm Outer Ring VD	-0.3293	0.0811
3 mm Circle VD	-0.4123	0.0262*
3 mm Ring VD	-0.4812	0.0082*

* P value significant as < 0.05

OCTA=optical coherence tomography; OCTA=optical coherence tomography angiography; MRI=magnetic resonance imaging; CST=central subfield thickness; CVI=choroidal vascularity index; GC-IPL=ganglion cell- inner plexiform layer; RNFL=retinal nerve fiber layer; FAZ=foveal avascular zone; PD=perfusion density; VD=vessel density

There were no significant correlations between GC-IPL thickness, RNFL thickness, or CST with any volumetric MRI parameters. Our findings contrasted with previous studies reporting significant associations between OCT

and MRI findings in patients with AD and MCI. Zhao et al. reported that RNFL thinning significantly correlated with reduced hippocampal volume, which is a contributing factor to ventricular expansion, in a cohort of both

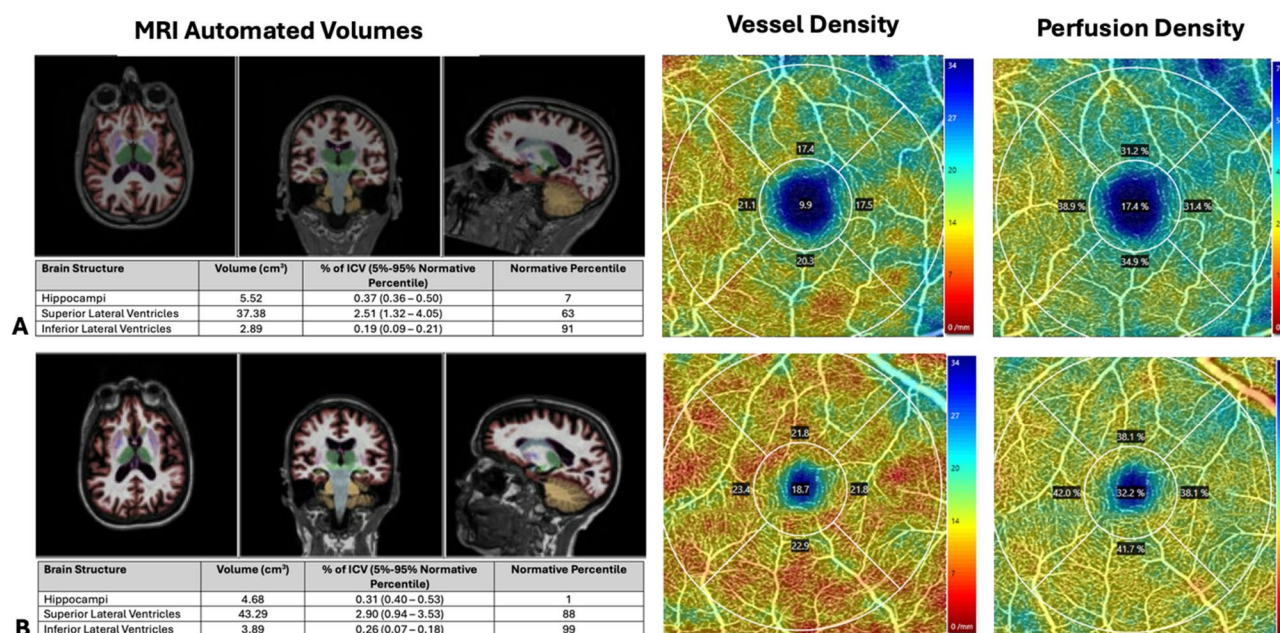


Fig. 1 Volumetric magnetic resonance imaging (MRI) images (left column) from NeuroQuant Age Related Atrophy Reports with normalized brain volumes and corresponding 3 mm x 3 mm optical coherence tomography angiography (OCTA) images of the superficial capillary plexus (with 3 mm circles superimposed) showing vessel density (VD) and perfusion density (PD) maps of the right eye of two participants, one with Alzheimer's disease (AD) and one with amnesic mild cognitive impairment (MCI). Scale on the right of the OCTA images shows blue indicating preserved VD and PD and red indicating loss of VD and PD. **(A)** A 69-year-old with amnesic MCI. The inferior lateral ventricle volume (ILV) is 2.89 cm³, which is at the 91st percentile of an age-matched reference. The superior lateral ventricle (SLV) volume is 37.39 cm³, which is at the 63rd percentile of reference. The hippocampal volume is 5.52 cm³, which is at the 7th percentile of reference. **(B)** A 65-year-old female with AD. The inferior lateral ventricle volume (ILV) is 3.89 cm³, which is at the 99th percentile of an age-matched reference. The superior lateral ventricle (SLV) volume is 43.29 cm³, which is at the 88th percentile of reference. The hippocampal volume is 4.68 cm³, which is 1st percentile of reference. There is reduction in both VD and PD in the patient with AD compared to the patient with amnesic MCI

AD and MCI patients [40]. Additionally, a retrospective cross-sectional study by Galvin et al. reported a direct correlation between hippocampal volume and GC-IPL thickness in AD and MCI patients [41]. Similarly, Ong et al. observed that GC-IPL thinning correlated with decreases in temporal lobe volume in individuals with cognitive impairment without dementia [42]. There is also research showing that GC-IPL thinning is related to decreased temporal lobe volume and whole brain atrophy, which can lead to ventricular expansion [43]. However, the findings from our study suggest that targeted analysis of retinal microvasculature through OCTA may be more sensitive to changes to brain volumetry in AD and aMCI compared to OCT structural parameters. Larger cohorts are needed to better understand the relationship between volumetric brain MRI findings and OCT structural parameters in neurocognitive disease.

In this study, CVI did not significantly correlate with any volumetric MRI parameters in either the aMCI or AD cohorts. Previously, our group found that CVI was significantly lower in MCI patients compared to controls, while both TCA and LA were significantly greater in patients with AD and MCI compared to controls [44]. More recently, Burke et al. reported no significant

difference in CVI between APOE $\epsilon 4$ carriers and non-carriers; however, total choroidal area and total vessel area were significantly larger in APOE $\epsilon 4$ carriers [45]. Kwapong et al. group found significantly lower choriocapillaris density on OCTA of AD patients compared to controls [46]. Another paper found that choroidal thickness was significantly thinner in AD patients compared to controls and also found a positive correlation between hippocampal volume and subfoveal choroidal thickness (SFCT), suggesting that as hippocampal volume decreases, the choroid also thins [47]. Additionally, multiple groups have observed choroidal thinning in AD patients relative to controls [48, 49, 50]. While structural changes such as choroidal thickness have been associated with neurocognitive disease, further work is needed to understand the relationship between CVI and structural brain MRI changes in aMCI and AD.

We also observed significant differences in OCT, OCTA, and volumetric MRI parameters when comparing directly between AD and aMCI cohorts. GC-IPL thickness was significantly reduced in AD patients compared to aMCI patients, consistent with prior work demonstrating more pronounced inner retinal thinning in Alzheimer's disease relative to MCI [51, 52]. Vessel

density in the 3 mm circle and ring was also significantly lower in AD patients. In our previous OCTA study, Yoon et al. reported decreased macular vessel density in AD compared to MCI, suggesting progressive retinal microvascular compromise with disease severity [36]. Additionally, ILV volume was significantly greater in the AD group relative to aMCI, mirroring established patterns of ventricular expansion associated with the transition from MCI to AD [24, 25, 37]. These findings reinforce the potential of both retinal and cerebral imaging biomarkers to differentiate clinical stages along the continuum of neurocognitive disease.

The validation of retinal biomarkers in AD hinges on the pathophysiologic importance of these changes. While many have claimed that the eye is a “window to the brain,” this statement is still being explored [53]. Cerebral neurodegeneration is also likely accompanied by loss of cerebral microvasculature [54, 55, 56]. Den Haan and colleagues suggested that macular thickness and RNFL thickness did not discriminate individuals with early-onset AD from controls; however, retinal thickness changes were significantly correlated with parietal cortical atrophy on MRI [57]. Bench research has also shown that retinal thickness and brain atrophy occur in concert in a mouse model of AD [58]. Our group as well as other investigators have previously shown that the retinal microvasculature differs between individuals with AD, MCI, and controls, and in this study, we show that these microvascular changes correlate with cerebral atrophy and increased ventricular volume on MRI [12].

This study has several limitations. As a cross-sectional study, we are unable to assess differences in rate of change over time, which could be an important diagnostic and prognostic factor in aMCI and AD. Additionally, patients with aMCI and AD were diagnosed by expert neurologists based on consensus clinical guidelines; however, we did not have amyloid biomarkers available for all patients. Thus, we cannot definitively state which aMCI patients had preclinical AD (i.e. aMCI with positive amyloid biomarkers suggestive of underlying AD pathology) and which AD patients had definitive evidence of underlying amyloid deposition. Future research integrating CSF studies and amyloid PET could provide a more direct measurement of disease burden.

Furthermore, brain imaging markers, such as the volume of FLAIR white matter hyperintensities or Fazekas score, may help us understand the contribution of cerebrovascular pathology. Additionally, the results of our study may not be generalizable to imaging techniques and analysis formats that differ from NeuroQuant, as brain volumes may be analyzed and segmented differently using other analysis protocols [59]. In our study, hippocampal volume was selected for analysis due to its role as a strong predictor of progression from mild

cognitive impairment (MCI) to Alzheimer's disease (AD) [18]. SLV and ILV volume were chosen because ventricle volume was identified as a potential variable associated with retinal parameters in our prior pilot analysis [19]. However, we acknowledge that future studies could incorporate a broader range of brain regions to fully capture potential neurodegenerative changes.

Of note, retinal and choroidal measurements may be subject to diurnal variations [60]. Previous studies have also documented that intraocular pressure, axial length, and refractive error may influence retinal and choroidal measurements in OCT and OCTA [61, 62, 63]. However, CVI used in our study is less likely to be affected than choroidal thickness measurements [64, 65]. We excluded eyes with any history of glaucoma as well as those with a refractive error $\pm$ 6 diopters to minimize potential effects of axial length. In a previous study of eyes with spherical equivalent between -8.0 D and $+5.0$ D and axial length of 23.29 to 24.33 mm, Sampson et al. found that there was less than 5% change in macular OCTA measurements [66]. Additionally, while patients with cataracts and previous history of cataract surgery were included in this study, reduced OCT signal strength has been demonstrated in individuals with significant lens opacity [67]. We excluded eyes with corrected VA worse than 20/40 on day of image acquisition, thus minimizing the imaging of eyes with significant lens opacity. Although we did not control for these variables in the study, they could be accounted for in future research. Our study contains a relatively small sample size, which means that our results may not be generalizable to all individuals with aMCI and AD. Continued work using larger and more diverse cohorts with known amyloid status will better support our findings that retinal vascular and structural changes are related to the pathophysiology of AD. There was no control group since cognitively normal individuals rarely undergo volumetric MRI imaging. Finally, future studies that perform more targeted analysis should correct for multiple comparisons, such as Bonferroni correction.

We report the largest study to date assessing correlations between retinal microvascular and structural parameters and volumetric MRI parameters in aMCI and AD. We find that OCTA VD and PD are related to hippocampal, SLV, and ILV volumes in individuals with aMCI and AD. These findings suggest a pathophysiologic link between retinal microvascular changes and AD-related cerebral neurodegeneration, warranting further investigation into the etiology and implication of these changes.

Abbreviations

AFI	Adjusted flow index
AD	Alzheimer's disease
aMCI	Amnesic mild cognitive impairment
CSF	Cerebrospinal fluid

CST	Central subfield thickness
CVI	Choroidal vascularity index
COIN	Comprehensive Ocular Imaging Network
EDI	Enhanced depth imaging
ETDRS	Early Treatment Diabetic Retinopathy Study
FAZ	Foveal avascular zone
GC-IPL	Ganglion cell-inner plexiform layer
ILV	Inferior lateral ventricle
LA	Luminal area
MMSE	Mini mental state exam
MRI	Magnetic resonance imaging
OCT	Optical coherence tomography
OCTA	Optical coherence tomography angiography
PET	Positron emission tomography
PD	Perfusion density
RNFL	Retinal nerve fiber layer
SCP	Superficial capillary plexus
SFCT	Subfoveal choroidal thickness
SLV	Superior lateral ventricle
TCA	Total choroidal area
VD	Vessel density

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Author contributions

Author's Contributions: WZ: conceptualization, investigation, formal analysis, writing, project administration, CBR: conceptualization, methodology, investigation, writing, project administration, DSG: writing, supervision, project administration, HP: investigation, formal analysis, SS: investigation, AJL: resources, KGJ: resources, JRP: resources, DP: resources, SSS: formal analysis, RA: software, writing, SF: conceptualization, writing, supervision, project administration, funding acquisition. All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to patient confidentiality but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study (NCT03233646) was reviewed and approved by the Duke University Health System Institutional Review Board (Pro00111831). Written informed consent was obtained from all participants and/or their designated legally authorized representative prior to study enrollment. This study adhered to the tenets of the Declaration of Helsinki with regard to enrollment of human subjects.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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